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π from Pure Arithmetic: A Spectral Phase Transition in the Binary Carry

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π from Pure Arithmetic: A Spectral Phase Transition in the Binary Carry Bridge

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Abstract

We study the propagation of arithmetic carries in binary multiplication and discover a spectral phase transition that converts rational arithmetic into a transcendental constant. Given two uniformly random K -bit integers X, Y and restricting to the D -odd pairs for which the product $P = XY$ is written in $D = 2K - 1$ bits, define the *sector ratio* $R(K) = \sigma_{10}(K)/\sigma_{00}(K)$, where σ_{ac} is the carry-weighted partition function restricted to pairs whose second-highest bits of X, Y are (a, c) .

We prove:

1. The Markov (independent-convolution) model gives a baseline $R_{\text{Markov}}(L) = S(\beta = 1, L)$ tending to $+2/3$ with explicit finite-size correction (Theorem 2).
2. The alternating spectral sum over the Dirichlet bridge has the exact closed form $S(\beta) = 2\beta/(1 + 2\beta)$, a Möbius transformation (Theorem 1).
3. Setting $S = -\pi$ yields the critical parameter $A^* = (3\pi + 2)/(2(\pi + 1))$, placing the system at distance $1/(2(\pi + 1))$ from the pole (Corollary 2).
4. The spectral zeta function of the Markov carry bridge reproduces $\zeta(2s)$ at all even arguments of the Riemann zeta function, via Weyl asymptotics (Theorem 3).
5. No D -odd pair has sector $(1,1)$ (Theorem 4), and the count ratio $N_{10}/N_{00} \rightarrow (2\ln(4/3) - 1/2)/(2\ln(9/8)) = 0.31993\dots$ involves only $\ln 2$ and $\ln 3$ (Theorem 5). The entire factor π in R enters exclusively through the carry weight ratio $\langle \text{val} \rangle_{10}/\langle \text{val} \rangle_{00}$.

We conjecture:

6. The exact cascade sector ratio satisfies $R(\infty) = \lim_{K \rightarrow \infty} R(K) = -\pi = -4 L(1, \chi_4)$ (Conjecture 1). Exact enumeration up to $K = 21$ (1.1×10^{12} digit pairs) confirms this to 4.0 significant digits by exponential Richardson extrapolation (§9.2; 4.4 digits using exact values only). The weight $\sin(n\pi/2)$ in the spectral sum equals the Dirichlet character $\chi_4(n)$, which emerges as the effective spectral weight under resolvent universality [E]. In the companion carry-Dirichlet program [L], the same stopping-time channel is realized as a canonical weighted first-return resolvent and reduces the remaining $s = 1$ closure to a scalar $\chi_4 \rightarrow L(1, \chi_4)$ constant together with preservation of the conditioned $1/2$ rate.

The system undergoes a spectral phase transition: the Markov model (subcritical) produces the rational $+2/3$; digit correlations push the carry bridge past a critical point into a supercritical regime where a transcendental value numerically consistent with $-\pi$ is observed. Combined with $\zeta(2s)$ from the spectral zeta function and the conjectural $L(1, \chi_4)$ contribution from the sector ratio, the carry bridge suggests a connection to both factors of the Dedekind zeta function $\zeta_{\mathbb{Q}(i)}(s) = \zeta(s) \cdot L(s, \chi_4)$ of the Gaussian integers.

1. Introduction

The appearance of π in number theory is usually traced to analytic methods: the residue calculus in the proof of the prime number theorem, the Fourier analysis underlying Dirichlet L-functions, or the Gaussian integral in the central limit theorem. In each case, the geometric content of π (circumference, area, rotation) enters through an analytical tool applied to a discrete problem.

In this paper we exhibit a mechanism that produces π from *purely arithmetic operations* — binary multiplication and carry propagation — without invoking any geometric or analytic machinery except at the final step of evaluating a closed-form identity. The mechanism reveals a spectral phase transition: a rational baseline asymptotic to $+2/3$ is converted into a transcendental one ($-\pi$) when digit correlations push the system past a critical point.

1.1 A First Look: Sectors, Carries, and π

Before the formal development, we illustrate the phenomenon with a concrete example.

The setup. Take all 4-bit binary numbers: $X, Y \in \{8, 9, \dots, 15\}$. Their products range from $8 \times 8 = 64$ to $15 \times 15 = 225$. Some products have 7 binary digits ($64 \leq P < 128$), others have 8 ($128 \leq P \leq 225$). We focus on the **7-bit products** — the “D-odd” pairs, where the product has exactly $D = 2K - 1 = 7$ digits. This is a natural boundary condition: the carry chain starts and ends at zero.

The sector partition. Each 4-bit number has the form $1abc_2$. The second-highest bit a splits the numbers into two groups: $\{8, 9, 10, 11\}$ (second bit = 0) and $\{12, 13, 14, 15\}$ (second bit = 1). The pair (X, Y) falls into one of four **sectors** based on these bits:

	Y: second bit = 0 {8, 9, 10, 11}	Y: second bit = 1 {12, 13, 14, 15}
X = 0 {8–11}	Sector (0,0) 16 pairs ALL D-odd	Sector (0,1) 16 pairs 8 of 16 D-odd
X = 1 {12–15}	Sector (1,0) 8 of 16 D-odd	Sector (1,1) ■ 0 D-odd ■ all products have 8 bits!

A striking fact: **sector (1,1) has zero D-odd pairs** (Theorem 4). When both factors have a 1 in the second-highest position, the product always overflows to 8 bits: $12 \times 12 = 144 \geq 128$. This is a theorem, not a coincidence — it holds for all K .

The carry weight. For each D-odd pair, we compute the carry chain (see [A] for the full spectral theory of carries). The *cascade valuation* scans the carry chain from the top and reads off a weight of $+1$, 0 , or -1 depending on the carry value at the

stopping position (formally, $w_{\text{casc}} = c_{M-1} - 1$). Summing these weights across all D-odd pairs in each sector gives the sector partition functions σ_{00} and σ_{10} . Their ratio $R(K) = \sigma_{10}/\sigma_{00}$ is the **sector ratio**.

The punchline. Computing $R(K)$ by exhaustive enumeration for increasing K :

K	Pairs enumerated	$R(K)$	$ R(K) + \pi $
7	$4^6 = 4,096$	-0.09	3.05
11	$4^{10} \approx 10^6$	-1.55	1.59
15	$4^{14} \approx 2.7 \times 10^8$	-2.82	0.32
19	$4^{18} \approx 6.9 \times 10^{10}$	-3.110	0.032
21	$4^{20} \approx 1.1 \times 10^{12}$	-3.133	0.008
∞ (extrapolated)		$-3.1416 \dots$	$< 10^{-3}$

Numerically, the ratio appears to converge to $-\pi = -3.14159 \dots$, confirmed to 4.0 significant digits by Richardson extrapolation (§9.2). The number π — the ratio of a circle’s circumference to its diameter — thus appears to emerge from *counting carry patterns in binary multiplication*.

The Markov model (independent carries) predicts a baseline tending to $R = +2/3$. The observed value points toward $-\pi$. The sign is wrong, the magnitude is wrong, and the answer is transcendental instead of rational. Digit correlations — the fact that adjacent carries share input bits — drive the system through a spectral phase transition (§5) from the rational Markov baseline to the transcendental limit.

1.2 Overview of Results

We also show that the Markov transfer operator for the carry bridge, viewed as a spectral zeta function, reproduces $\zeta(2s)$ at all even arguments — a manifestation of Weyl’s law in the arithmetic setting. Combined with the conjectural $L(1, \chi_4) = \pi/4$ from the sector ratio, this connects the carry bridge to the Dedekind zeta function of the Gaussian integers.

Scope and status. Seven theorems are proved unconditionally (Theorems 1–7). The central claim $R(\infty) = -\pi$ (Conjecture 1) is supported by 4.0-digit numerical evidence (exponential Richardson, §9.2) but remains open; under the Linear Mix Hypothesis of [E], it follows from the proved closed-form identity $S(A) = 2(1-A)/(3-2A)$. This paper is therefore a contribution to experimental mathematics: it identifies the phenomenon, proves the structural framework, and isolates the remaining closure problem. The remaining proof targets are: (i) prove the scalar identity $C = -4$ in $R(\infty) = C \cdot L(1, \chi_4)$, and (ii) prove analytically that the conditioned D-odd chain preserves the dominant $1/2$ spectral rate. The stopping-time framework extends to a full Dirichlet series in the complex variable s in $[L]$, where the mixed channel is realized as a canonical weighted first-return resolvent and exhibits corrected- L^2 structure on $\text{Re}(s) > 1$; the corresponding carry-side object remains zero-free in the tested critical-strip box.

1.3 Setting

Let X and Y be independent uniformly random integers in $[2^{\{K-1\}}, 2^K)$ (K -bit integers with leading bit 1). On the D -odd subdomain studied here, their product $P = XY$ has $D = 2K - 1$ binary digits. The schoolbook multiplication algorithm computes

$$P = \sum_{j=0}^{D-1} p_j \cdot 2^j$$

where at each position j the product digit and carry are determined by

$$p_j = (\text{conv}_j + c_j) \bmod 2, \quad c_{\{j+1\}} = \lfloor (\text{conv}_j + c_j) / 2 \rfloor$$

with $\text{conv}_j = \sum_i x_i \cdot y_{\{j-i\}}$ the convolution at position j and $c_0 = 0$ the boundary condition.

Definition 1 (Sector partition). The *sector* of a pair (X, Y) is $(a, c) \in \{0,1\}^2$ where $a = x_{\{K-2\}}$ and $c = y_{\{K-2\}}$ are the second-highest bits. The *sector partition function* is

$$\sigma_{ac}(K) = \sum_{(X,Y): x_{\{K-2\}}=a, y_{\{K-2\}}=c} w(X, Y)$$

where $w(X, Y) = w_{\text{casc}}(X, Y)$ is the cascade carry weight defined below. Two valuations appear in the literature: the *schoolbook* weight $w_{\text{sch}} = 2c_{D-2} - 1$ (fixed-position readout) and the *cascade* weight $w_{\text{casc}} = c_{M-1} - 1$ where M is the topmost nonzero carry position (first-passage readout; not to be confused with the companion matrix M of $[A], [B]$). Throughout this paper $w = w_{\text{casc}}$ unless otherwise noted; it is this valuation that produces $-\pi$.

Definition 2 (Sector ratio). $R(K) = \sigma_{10}(K)/\sigma_{00}(K)$. By symmetry, $\sigma_{10} = \sigma_{01}$. (Not to be confused with the carry correction factor $R(l, s)$ of $[B]$, which measures the per-prime deviation from the Euler product.)

1.4 Main results

Theorem 1 (Spectral closed form). For $\beta \in (-1/2, +\infty)$, define

$$S(\beta, L) = \sum_{k=0}^{\lfloor L/2 \rfloor - 1} (-1)^k / (1/2 + \beta \sin^2((2k+1)\pi/(2L)))$$

Then

$$\lim_{L \rightarrow \infty} S(\beta, L) = 2\beta/(1 + 2\beta)$$

Corollary 1 (Möbius form). Setting $\beta = 1 - A$, the sum as a function of the eigenvalue shift A is

$$S(A) = 2(1-A)/(3-2A)$$

a Möbius transformation with a zero at $A = 1$ and a pole at $A = 3/2$.

Corollary 2 (Phase transition).

- $A = 0$ (Markov): $S = +2/3$
- $A = 1$ (critical): $S = 0$
- $A = A$: $S = -\pi$, where $A = (3\pi + 2)/(2(\pi + 1)) = 3/2 - 1/(2(\pi+1)) \approx 1.37927$

Theorem 2 (Markov baseline). The independent-convolution (Markov) model gives $R_{\text{Markov}}(L) = S(\beta = 1, L) \rightarrow +2/3$, with finite-size correction (writing $S(\beta, L)$ as in Theorem 1 with $\beta = 1 - A$; the Markov case $A = 0$ corresponds to $\beta = 1$):

$$S(\beta = 1, L) = 2/3 + 10 \zeta(2)/(3 L^2) + O(L^{-4})$$

Theorem 3 (Spectral zeta function). Let $F = \frac{1}{2}I - T_{\text{bridge}}$ be the fluctuation operator of the Markov carry bridge of length L , with eigenvalues $\varepsilon_n = \sin^2(n\pi/(2L))$. Then for $\text{Re}(s) > 1/2$,

$$\zeta_F(s) / (2L/\pi)^{2s} \rightarrow \zeta(2s) \quad \text{as } L \rightarrow \infty$$

with convergence rate $O(L^{1-2\sigma})$ where $\sigma = \Re(s)$ (reducing to $O(1/L^2)$ for $\sigma > 3/2$).

Theorem 4 ($N_{11} = 0$). No D-odd pair has sector (1,1).

Theorem 5 (Closed-form count ratio). $N_{10}/N_{00} \rightarrow (2\ln(4/3) - 1/2)/(2\ln(9/8)) = 0.31993\dots$, involving only $\ln 2$ and $\ln 3$. The convergence correction is $O(1/2^K)$.

Conjecture 1 (Exact sector ratio). $R(\infty) = \lim_{K \rightarrow \infty} \sigma_{10}(K)/\sigma_{00}(K) = -\pi$. (Note: this is implied by, but weaker than, the Linear Mix Hypothesis of [E], which is also called “Conjecture 1” in that paper.)

Supported by exact enumeration to $K = 21$ with Richardson extrapolation (Section 9). By the decomposition $R = R_0 + \Delta R$ [E], the conjecture reduces to proving that the cascade correction $\Delta R = \text{Tr}[(I-T)^{-1} \cdot P_{\{c=0\}}]$ equals exactly $-\pi - R_0 = +0.7896\dots$

1.5 Structure of the paper

- Section 2: Carry bridge and Dirichlet eigenfunctions
- Section 3: Markov baseline ($R \rightarrow +2/3$)
- Section 4: Proof of the closed form (Theorem 1)
- Section 5: The spectral phase transition
- Section 6: The Dirichlet character χ_4 and $L(1, \chi_4)$
- Section 7: The spectral zeta function and $\zeta(2s)$ (Theorem 3)
- Section 8: The Bernoulli hierarchy and the Gaussian integers
- Section 9: Numerical evidence
- Section 10: The gap between $S(A)$ and $R(K)$
- Section 11: Open problems

2. The Carry Bridge and Dirichlet Eigenfunctions

2.1 The D-odd constraint

A pair (X, Y) with $X, Y \in [2^{\{K-1\}}, 2^K]$ has product P with exactly $D = 2K - 1$ bits (the “D-odd” constraint). This forces $c_0 = 0$ (no carry into position 0) and $c_{D-1} = 0$ (no overflow beyond $D - 1$ bits). The carry chain $c_0, c_1, \dots, c_{D-1}$ with these boundary conditions forms a *carry bridge* — a lattice path conditioned to return to zero, analogous to a Brownian bridge.

2.2 Transfer operator and eigenvalues

The Markov transfer operator for the carry chain, conditioned on independent convolutions at each position, was introduced by Diaconis and Fulman [1]. In the base-2 case, the carry at each position takes values in $\{0, 1\}$, and the transition matrix has eigenvalues

$$\lambda_n = (1/2) \cos(n\pi/L), \quad n = 1, \dots, L-1$$

where $L = D = 2K - 1$ is the bridge length. The corresponding eigenfunctions are the Dirichlet sine functions $\phi_n(j) = \sin(n\pi j/L)$, reflecting the zero boundary conditions $c_0 = c_L = 0$. The leading eigenvalue $\lambda_1 = (1/2)\cos(\pi/L) \rightarrow 1/2$ encodes the Diaconis-Fulman spectral gap.

2.3 Spectral representation of the sector response

The sector perturbation at position $j^* = K - 2$ excites the Dirichlet modes of the carry bridge. The response at the boundary is a spectral sum

$$R \sim \sum_{n=1}^{L-1} \sin(n\pi j^*/L) / (1 - \lambda_n^{\text{eff}})$$

where λ_n^{eff} incorporates both the Markov eigenvalue and the effective modification due to digit correlations. In the Markov model, the perturbation position satisfies $j/L = (K-2)/(2K-1) \rightarrow 1/2$ as $K \rightarrow \infty$, giving $\sin(n\pi j/L) \rightarrow \sin(n\pi/2) = \chi_4(n)$, the non-principal Dirichlet character modulo 4 (§6). In the true cascade, the stopping position is not localized at the midpoint (it concentrates near the top of the chain; see [E, §8.5]). Nonetheless, under the LMH, the resolvent-weighted spectral sum converges to $-\pi$ through a collective mechanism [E, §8.3] that is functionally equivalent to the midpoint evaluation (§6.2).

The precise relationship between $R(K)$ and the spectral sum $S(A, L)$ is subtle: [E, §8.1–§8.3] establish that $S(A, L)$ is not a direct model of $R(K)$, but rather that both quantities converge (numerically) toward $-\pi$ through different mechanisms (§10). Exact enumeration through $K = 21$ (5.5×10^{11} pairs; [E, Tables 1–2]) confirms convergence to $-\pi$ with 2.1 significant digits raw and 4.0 digits by exponential Richardson extrapolation (§9). [E] identifies the mechanism as **resolvent universality** of the effective transfer operator. Formalizing this as a rigorous identity remains the principal open problem.

3. The Markov Baseline: $R \rightarrow +2/3$

3.1 Independent convolutions

In the Markov model, convolutions at distinct positions are treated as independent random variables given the carry state. The sector perturbation modifies the convolution distribution at position $K - 2$ by conditioning on the second-highest bits (a, c) of the input.

3.2 Proof of the Markov Baseline

With $A = 0$ (no correlation-induced shift), the spectral sum reduces to

$$S(\beta = 1, L) = \sum_{n \text{ odd}} (-1)^{(n-1)/2} / (1 - (1/2)\cos(n\pi/L))$$

This is exactly the Markov case $S(\beta = 1, L)$ in the notation of Theorem 1, giving $S(\beta = 1) = 2 \cdot 1/(1 + 2) = 2/3$.

Proposition 1 (Finite-size correction).

$$S(\beta = 1, L) - 2/3 = 5\pi^2/(9L^2) + O(L^{-4}) = 10 \zeta(2)/(3L^2) + O(L^{-4})$$

The correction coefficient $5\pi^2/9$ is derived analytically from the Fourier expansion (Section 4) and verified numerically to 6 significant digits.

4. Proof of the Closed Form

Proof of Theorem 1. The proof proceeds in five steps.

Step 1: Fourier expansion. The integrand has the standard Fourier series (see e.g. [10])

$$1/(1/2 + \beta \sin^2 t) = (2/\gamma) [1 + 2 \sum_{m=1}^{\infty} r^m \cos(2mt)]$$

where $\gamma = \sqrt{1 + 2\beta}$ and $r = (\gamma - 1)^2/(2\beta)$, valid for $|r| < 1$ (equivalently $\beta > -1/2$). This follows from the identity for $1/(A - B \cos \theta)$ with $A = (1 + \beta)$, $B = \beta$, $\theta = 2t$.

Step 2: Alternating midpoint sum. For $L \equiv 0 \pmod{4}$, the alternating midpoint sum satisfies

$$T_m(L) := \sum_{k=0}^{L/2-1} (-1)^k \cos(m(2k+1)\pi/L) = \begin{cases} 1/\cos(m\pi/L) & \text{if } m \text{ is odd,} \\ 0 & \text{if } m \text{ is even.} \end{cases}$$

Proof. By geometric summation, $\sum_k (-1)^k e^{im(2k+1)\pi/L} = e^{im\pi/L} (1 - (-1)^{L/2+m}) / (1 + e^{2im\pi/L})$. When $L/2$ is even and m is odd, the numerator is 2 and the denominator is $2\cos(m\pi/L)$. ■

Step 3: Assembly. Substituting the Fourier expansion into $S(\beta, L)$:

$$\begin{aligned} S(\beta, L) &= (2/\gamma) [T_0 + 2 \sum_{m=1}^{\infty} r^m T_m(L)] \\ &= (4/\gamma) \sum_{j=0}^{\infty} r^{2j+1} / \cos((2j+1)\pi/L) \end{aligned}$$

($T_0 = 0$ because the alternating sum of a constant vanishes when $L/2$ is even.)

Step 4: Continuum limit. As $L \rightarrow \infty$, $\cos((2j+1)\pi/L) \rightarrow 1$ for each fixed j . Since $|r| < 1$, dominated convergence gives

$$S(\beta, \infty) = (4r/\gamma) \cdot 1/(1 - r^2)$$

Step 5: Simplification. Direct computation using $r = (\gamma - 1)^2/(2\beta)$ and $\gamma^2 = 1 + 2\beta$:

$$4r / (\gamma(1 - r^2)) = 4 \cdot (\gamma - 1)^2 / (2\beta) / (\gamma \cdot 4\gamma(\gamma - 1) / (2\beta)^2)$$

After cancellation, this simplifies to

$$S(\beta, \infty) = 2\beta / (1 + 2\beta) \quad \blacksquare$$

Proposition 2 (General correction coefficient). The $O(1/L^2)$ correction to $S(\beta, L)$ is

$$S(\beta, L) = 2\beta / (1 + 2\beta) + C(\beta)/L^2 + O(L^{-4})$$

where

$$C(\beta) = 2\pi^2 r(1 + 6r^2 + r^4) / (\gamma(1 - r^2)^3)$$

For $\beta = 1$: $C(1) = 5\pi^2/9 = 10\zeta(2)/3$, confirming Proposition 1.

Proof. Expand $1/\cos((2j+1)\pi/L) = 1 + (2j+1)^2\pi^2/(2L^2) + O(L^{-4})$, substitute into Step 3, and sum the geometric series. ■

Remark (M-parity subtlety). The closed form $S(\beta, \infty) = 2\beta/(1+2\beta)$ applies when $M = \lfloor L/2 \rfloor$ is even. For M odd, the limit differs by an $O(1)$ term at finite L , but both subsequences converge to the same limit. This is because the additional term vanishes as $L \rightarrow \infty$.

5. The Spectral Phase Transition

5.1 Interpretation

The closed form $S(A) = 2(1-A)/(3-2A)$ reveals a *spectral phase transition* parameterized by the eigenvalue shift A :

A	S(A)	Regime
0	+2/3	Subcritical (Markov): rational
1/2	+1/2	Subcritical
1	0	Critical: exact cancellation
1.2	-2/3	Supercritical
$A^* \approx 1.379$	$-\pi$	Supercritical: transcendental
3/2	$\pm\infty$	Pole (divergence)

5.2 The critical point

At $A = 1$ ($\beta = 0$), all effective eigenvalues equal $1/2$, and every term in the alternating sum has the same magnitude. The sum vanishes by exact cancellation: $\sum (-1)^k \cdot 2 = 0$. This is the boundary between the subcritical (positive) and supercritical (negative) regimes.

5.3 Subcritical vs. supercritical

For $A < 1$, the effective eigenvalues are below $1/2$ and the alternating sum is positive (dominated by the $k = 0$ term). For $A > 1$, eigenvalues exceed $1/2$, denominators shrink, and the negative terms dominate. The sign flip at $A = 1$ is the *trace anomaly*: digit correlations reverse the sign of the classical (Markov) result.

5.4 Distance to the pole

The value $A^* = 3/2 - 1/(2(\pi+1))$ places the physical system at distance $1/(2(\pi+1)) \approx 0.121$ from the pole of the Möbius transformation. The formula is self-referential: A^* contains π , and $S(A^*) = -\pi$. This is not circular — it is a well-defined algebraic relation $2(1-A)/(3-2A) = -\pi$ with a unique solution $A = (3\pi+2)/(2(\pi+1))$.

6. The Dirichlet Character and L-function

6.1 $\sin(n\pi/2) = \chi_4(n)$

The weight function in the spectral sum takes the values

$$\begin{aligned}\sin(n\pi/2) &= +1 \text{ if } n \equiv 1 \pmod{4}, \\ &= -1 \text{ if } n \equiv 3 \pmod{4}, \\ &= 0 \text{ if } n \text{ is even,}\end{aligned}$$

which is exactly the non-principal Dirichlet character $\chi_4: (\mathbb{Z}/4\mathbb{Z})^* \rightarrow \{\pm 1\}$.

6.2 The effective χ_4 model

In the Markov (independent-convolution) model, the sector perturbation acts at position $j^* = K - 2$, and the bridge has length $L = 2K - 1$. Since $j/L \rightarrow 1/2$ as $K \rightarrow \infty$, the Dirichlet mode evaluation $\sin(n\pi j/L) \rightarrow \sin(n\pi/2) = \chi_4(n)$. This “midpoint selection” provides a clean motivation for why χ_4 appears in the spectral sum.

Caveat ([E]). In the true (non-Markov) cascade, the stopping position concentrates near the top of the chain (depth $j = 2, 3, 4$ from position $D-2$), not at the midpoint. The physical cascade stopping profile, projected onto the Dirichlet sine basis, does not equal $\chi_4(n)$ at finite K [E, §8]. Nonetheless, the resolvent

$$(I - \mathcal{K}_{\text{eff}})^{-1}$$

acts as a matched spectral filter that transmutes the true erratic cascade profiles into a macroscopic sum converging (under LMH) to $-\pi$ [E, §8.3]. The χ_4 model is therefore best understood as an *effective toy reward* — a mathematically tractable proxy that belongs to the same universality class as the true cascade in the thermodynamic limit.

6.3 Connection to $L(1, \chi_4)$

The Dirichlet L-function at $s = 1$ gives the Leibniz formula:

$$L(1, \chi_4) = \sum_{n=0}^{\infty} (-1)^n / (2n+1) = 1 - 1/3 + 1/5 - \dots = \pi/4$$

If $R = -\pi$ (Conjecture 1), then $R = -4 L(1, \chi_4)$. The carry bridge produces π through the same Dirichlet series that appears in the classical evaluation of π via the Leibniz–Gregory formula.

6.4 Structural identity

Combined with the sub-sector ratio $\alpha \rightarrow 1/6 = B_2$ (the second Bernoulli number) and $\zeta(2) = \pi^2/6$:

$$\alpha \cdot R^2 = (1/6) \cdot \pi^2 = \pi^2/6 = \zeta(2)$$

If both $\alpha = 1/6$ (Conjecture 2, §8) and $R = -\pi$ (Conjecture 1) can be established from carry arithmetic, this would yield a new proof of the Basel problem — one that begins with integer multiplication. **Caveat:** this depends on two currently unproved conjectures.

7. The Spectral Zeta Function and $\zeta(2s)$

7.1 The fluctuation operator

Definition 3 (Fluctuation operator). $F = \frac{1}{2}I - T_{\text{bridge}}$, where T_{bridge} is the Markov carry bridge transfer matrix. The eigenvalues of F are

$$\varepsilon_n = 1/2 - \lambda_n = \sin^2(n\pi/(2L)), \quad n = 1, \dots, L-1$$

Definition 4 (Spectral zeta function).

$$\zeta_F(s) = \sum_{n=1}^{L-1} \varepsilon_n^{-s} = \sum_{n=1}^{L-1} \sin^{-2s}(n\pi/(2L))$$

7.2 Proof of Theorem 3

For $n \ll L$, the small-angle approximation $\sin(x) \approx x$ gives $\varepsilon_n \approx (n\pi/(2L))^2$, so

$$\zeta_F(s) \approx (2L/\pi)^{2s} \sum_{n=1}^{L-1} n^{-2s}$$

We make this precise. Write $\varepsilon_n = (n\pi/(2L))^2 \cdot (\sin(x_n)/x_n)^2$ where $x_n = n\pi/(2L)$. Then

$$\varepsilon_n^{-s} = (2L/(n\pi))^{2s} \cdot (x_n/\sin(x_n))^{2s}$$

and

$$\zeta_F(s)/(2L/\pi)^{2s} = \sum_{n=1}^{L-1} n^{-2s} \cdot (x_n/\sin(x_n))^{2s}$$

Error analysis. The error has two sources:

(i) *Tail truncation.* The missing terms $\sum_{n \geq L} n^{-2s}$ contribute $O(L^{\{1-2s\}})$ for $\text{Re}(s) > 1/2$.

(ii) *Sine correction.* For each mode, $(x/\sin(x))^{2s} = 1 + sx^2/3 + O(x^4)$. Summing the leading correction:

$$\sum_{n=1}^{L-1} n^{-2s} \cdot s \cdot (n\pi/(2L))^2/3 = (s\pi^2/(12L^2)) \cdot \sum_{n=1}^{L-1} n^{2-2s}$$

For $s > 3/2$ this sum converges and the correction is $O(1/L^2)$. For $1/2 < s \leq 3/2$, the sum grows as $L^{\{3-2s\}}$ and the correction is $O(L^{\{1-2s\}}) = O(1/L)$ when $s = 1$.

The combined rate is $O(L^{1-2\sigma})$ for $1/2 < \sigma \leq 3/2$, improving to $O(1/L^2)$ for $\sigma > 3/2$.

■

Remark. This is a manifestation of Weyl's law [14]: for any self-adjoint operator on a one-dimensional domain of length L with Dirichlet boundary conditions, the spectral zeta function reproduces $\zeta(2s)$ through the asymptotic eigenvalue density at the spectral edge. The carry bridge operator is a concrete arithmetic realization of this universal phenomenon.

7.3 Numerical verification

Verified numerically for $s = 1, 2, 3, 4$ (even zeta values) and $s = 3/2, 5/2$ (odd zeta values $\zeta(3), \zeta(5)$):

s	$\zeta(2s)$	Ratio at L=100	Ratio at L=500	Ratio at L=1000
1	$\pi^2/6 = 1.6449$	1.6776	1.6515	1.6482
2	$\pi^4/90 = 1.0823$	1.1266	1.0910	1.0867
3/2	$\zeta(3) = 1.2021$	1.2291	1.2075	1.2048
5/2	$\zeta(5) = 1.0369$	1.0542	1.0404	1.0387

All ratios converge to the exact $\zeta(2s)$ values at rates consistent with Theorem 3 ($O(L^{1-2\sigma})$ for $\sigma \leq 3/2$, $O(1/L^2)$ for $\sigma > 3/2$).

7.4 Universality and non-universality

The $\zeta(2s)$ convergence is *universal*: it depends only on the Dirichlet boundary conditions and the length L , not on the specific transition probabilities. Any operator with the same boundary structure produces the same result.

By contrast, the conjectured limit $R \rightarrow -\pi$ is *not* universal — it depends on the binary carry structure, the Fejér kernel correlations, and the D-odd constraint (Section 10).

8. The Bernoulli Hierarchy and the Gaussian Integers

8.1 The Bernoulli hierarchy

The carry chain in binary multiplication encodes Bernoulli numbers at successive levels of its spectral decomposition:

Level	Carry quantity	Value	Bernoulli
0	Stationary distribution	1	$B_0 = 1$
1	Spectral gap λ_2	$1/2$	$ B_1 = 1/2$
2	Sub-sector ratio α	$1/6$	$B_2 = 1/6$
3	Value cancellation $\sigma_{00}^{\{c_2=1, c_3=1\}}$	0	$B_3 = 0$

Level 0 is trivial: the carry chain has a unique stationary distribution. **Level 1** is the Diaconis-Fulman theorem [1]: the spectral gap of the carry transfer operator in base b is $1/b$, giving $1/2 = |B_1|$ for binary. **Level 2** is new. The sub-sector ratio α measures the carry population at position $D-2$, which is directly linked to the boundary layer decomposition of [E, §8.5]: the $c_{D-2} = 1$ sub-population is precisely the “top” cascade contribution that is blocked in sector $(1,0)$.

Definition 5 (Sub-sector ratio). Define $\alpha(K) = \sigma_{00}^{\{c=1\}(K)} / \sigma_{00}^{\{c=0\}(K)}$, where $\sigma_{00}^{\{c=0\}}$ and $\sigma_{00}^{\{c=1\}}$ are the sector- $(0,0)$ partition function restricted to pairs with $c_{D-2} = 0$ and $c_{D-2} = 1$, respectively.

Conjecture 2 (Bernoulli sub-sector ratio). $\alpha(K) \rightarrow 1/6 = B_2$ as $K \rightarrow \infty$.

The evidence is preliminary: the overshoot at $K \geq 20$ means the conjecture rests on Richardson extrapolation (3.3 digits) rather than direct convergence. Additional data at higher K would strengthen the case.

Numerical evidence. Exact enumeration gives:

K	$\alpha(K)$	$ \alpha - 1/6 $	Sig. digits
7	0.15177	0.0149	1.2
10	0.15958	0.0071	1.5
13	0.16378	0.0029	1.9
16	0.16575	0.0009	2.4
18	0.16652	0.0001	2.7
20	0.16880	0.0021	2.0
21	0.16922	0.0026	2.6

The convergence is non-monotonic, with a two-phase structure: approach from below through $K = 18$, then overshoot at $K \geq 20$. The overshoot is present at $K = 21$ ($\alpha = 0.1692$, gap = 0.0026). Richardson extrapolation with the ansatz $\alpha(K) = \alpha(\infty) + a_1 K^2/2^K + \dots$ gives $\alpha(\infty) \approx 0.1667$ (3.3 digits), consistent with $1/6$. The correction structure is $K^2/2^K$ rather than pure $1/2^K$. The non-monotonic convergence is analogous to the oscillatory behaviour of R_{recon} in Paper E (§8.3), likely reflecting a Gibbs-type phenomenon in the spectral truncation.

8.2 The Euler-Maclaurin connection (speculative)

This connection is a structural analogy rather than a formal derivation; the numerical coincidence is striking but unproved.

The carry chain computes $\lfloor (\text{conv} + \text{carry})/2 \rfloor$ at each position — a discrete summation with rounding. The Euler-Maclaurin formula for the discrepancy between a discrete sum and its integral is

$$\Sigma f - \int f = B_1 \cdot \Delta f + B_2/2! \cdot \Delta f' + B_4/4! \cdot \Delta f''' + \dots$$

where B_n are Bernoulli numbers. The carry chain's spectral decomposition mirrors this hierarchy: - $B_1 = -1/2$ controls the first-order correction (exponential convergence rate = spectral gap); - $B_2 = 1/6$ controls the second-order correction (the sub-sector ratio α , which measures how much of the carry at $D - 2$ is “rescued” from the $c = 1$ sub-population).

This connection is structural rather than a formal derivation: the carry chain is a non-linear discrete dynamical system, not a trapezoidal sum. Nonetheless, the numerical evidence strongly supports the identification.

Theorem 6 (Cascade Rigidity). For D -odd sector- $(0,0)$ pairs, define the depth- j carry $c_j := \text{carries}[D - j]$ (the carry at position $D - j$, i.e., depth j from the MSB end). For $c_2 \in \{0, 1\}$:

- (i) $\text{carries}[D - 1] = \text{carries}[D] = 0$.
- (ii) If $c_2 = \text{carries}[D - 2] \geq 1$: $M = D - 2$ and $\text{val} = c_3 - 1$.
- (iii) If $c_2 = 0$ and $c_3 = \text{carries}[D - 3] \geq 1$: $M = D - 3$ and $\text{val} = c_4 - 1$.
- (iv) If $c_2 = c_3 = 0$: M depends on the full carry chain.

Proof. In sector $(0,0)$, $x_{K-2} = y_{K-2} = 0$ and $x_{K-1} = y_{K-1} = 1$. The convolution at position $D - 2$ is $\text{conv}_{D-2} = x_{K-2}y_{K-1} + x_{K-1}y_{K-2} = 0$. Then $\text{carries}[D - 1] = \lfloor (0 + c_2)/2 \rfloor = 0$ for $c_2 \in \{0, 1\}$, and $\text{carries}[D] = \lfloor (1 + 0)/2 \rfloor = 0$. Scanning from D downward, the first nonzero carry is at $D - 2$ when $c_2 \geq 1$, giving $M = D - 2$ and $\text{val} = \text{carries}[D - 3] - 1 = c_3 - 1$. When $c_2 = 0$, the scan continues to $D - 3$, etc. $\square$

The theorem extends recursively: at each level $n \geq 2$, if $c_n = \text{carries}[D - n] \geq 1$ while $c_2 = \dots = c_{n-1} = 0$, then $M = D - n$ and $\text{val} = c_{n+1} - 1$. This recursive extension is verified numerically through $n = 6$ at $K \leq 13$; a general inductive proof for all n is not given.

Corollary (Level 3: $B_3 = 0$). For $c_2 \geq 1$, Theorem 6(ii) gives $\text{val} = c_3 - 1$. When $c_3 = 1$, every pair contributes $\text{val} = 0$, so $\sigma_{00}^{c_2 \geq 1, c_3 = 1} = 0$ exactly. This is the carry-chain incarnation of $B_3 = 0$: the corresponding term in the Euler-Maclaurin formula vanishes. Verified for all $K \leq 13$ ($n = 1,789,942$ pairs at $K = 13$).

The recursive extension produces the same vanishing at every depth: $\sigma_{00}^{c_2=0, c_3 \geq 1, c_4=1} = 0$, and so on. At each level, only the “all-zeros” sub-population ($c_2 = c_3 = \dots = c_n = 0$) carries non-trivial cascade dynamics.

Level 4: open. The non-trivial spectral content telescopes into the $(c_2, c_3, c_4) = (0, 0, 0)$ and $(0, 0, 1)$ cells. Seven candidate partition-function ratios, the asymptotic correction structure of $\alpha(K)$, and the spectral gap of the near-boundary carry transfer matrix were tested through $K = 13$; none produces $|B_4| = 1/30$. Identifying the correct observable remains open.

8.2a Stopping-time decomposition

Theorem 6, combined with the unit leading carry (Theorem 7, §10.5), yields an exact stopping-time decomposition of the sector ratio:

$$\sigma_{ab} = \sum_{\tau=2}^{D-2} n_{ab}(\tau) \cdot E[\text{val} \mid \tau, ab]$$

where $\tau = D - M$ is the depth of the first nonzero carry from the MSB. The Cascade Rigidity Theorem makes this decomposition rigorous: at each τ , $\text{val} = c_{\tau+1} - 1$ where $c_{\tau+1} = \text{carries}[D - \tau - 1]$.

Structural exclusion. In sector $(1, 0)$, $\text{conv}_{D-2} = 1$ (Proposition 4), so $\text{carries}[D - 2] = 0$ and $P(\tau = 2 \mid \text{sector } 10) = 0$ for all K . Sector $(0, 0)$ has $\text{conv}_{D-2} = 0$, allowing $\tau = 2$.

Universality. Exact enumeration confirms that both $P(\tau \mid ab)$ and $E[\text{val} \mid \tau, ab]$ converge to K -independent constants.

Table 1. Stopping-time decomposition constants at $K = 21$ (from the $K = 21$ exact enumeration of $[E]$, 3.4×10^{11} D-odd pairs in the $X \leq Y$ half-plane); cross- K stability from $K = 19$ to $K = 21$ confirms convergence to 5-6 significant digits.

τ	$P(\tau \mid 00)$	$P(\tau \mid 10)$	$E[v \mid \tau, 00]$	$E[v \mid \tau, 10]$	$\Delta(\tau)$
2	0.54070	0.00000	-0.00812	—	—
3	0.13713	0.45726	-0.11761	+0.26971	+0.38732
4	0.10924	0.22875	-0.04089	+0.31022	+0.35111
5	0.08557	0.14398	-0.07338	+0.33738	+0.41076
6	0.05007	0.07849	-0.00867	+0.30657	+0.31525
7	0.03329	0.04308	-0.01270	+0.32311	+0.33580
8	0.01862	0.02313	+0.04186	+0.33033	+0.28848
9	0.01169	0.01231	+0.01742	+0.33307	+0.31565

The sector asymmetry $\Delta(\tau) = E[v \mid \tau, 10] - E[v \mid \tau, 00]$ is stable to < 0.00001 across $K = 19$ -21 for $\tau \leq 5$, confirming genuine universality. Richardson extrapolation (assuming convergence rate $\rho = 1/2$) yields 7-digit estimates of the limiting constants (experiment P1_06).

Identification status. PSLQ searches against $\{1, \pi, \ln 2, \ln 3\}$ on the Richardson-extrapolated limits produce candidate relations with residuals in the range 10^{-6} - 10^{-7} but no confident closed forms. The precision is sufficient to confirm universality but

not to prove algebraic identities. Extending the stopping-time analysis to $K \geq 24$ (computationally feasible with the C implementation of [E]) would provide the 10+ digits needed for reliable PSLQ identification.

8.2b Failure of the per-position spectral limit

A natural approach to the sector ratio is to analyze the per-position carry transition matrices $T(d)$ directly. Under D-odd conditioning, the carry chain at each position d induces a stochastic matrix on carry states $\{0, 1, 2, \dots\}$. In the Markov (independent-convolution) model, the bulk eigenvalue is $\lambda_2 = 1/2$ at every position; the Linear Mix Hypothesis [E] would predict $\lambda_2 = A^*/2 \approx 0.69$.

Empirical computation of the bulk-averaged transition matrix for $K = 5$ -12 (experiment P1_05) reveals a qualitatively different picture:

K	$\lambda_2(\text{bulk})$	A_{eff}
6	0.690	-3.25
8	0.756	-0.88
10	0.810	-0.40
12	0.858	-0.19

The second eigenvalue λ_2 grows monotonically toward 1, and the effective mixing parameter A_{eff} extracted from the LMH formula diverges to $-\infty$ instead of converging to $A^* \approx 1.38$. The per-position transition matrices become *more rigid* (closer to identity) as K increases — the D-odd conditioning progressively constrains the local transitions, which is the dynamical mechanism behind Theorem 6 (Cascade Rigidity).

Conclusion. The Linear Mix Hypothesis cannot hold at the per-position level. The LMH must describe a *macroscopic* property of the full resolvent

$$(I - \mathcal{K}_{\text{eff}})^{-1}$$

, not a uniform modification of each per-position factor. This is consistent with the resolvent universality mechanism identified in [E, §8.3]: the spectral resolvent achieves $-\pi$ through collective mode weighting, not through any single position's transition structure. The stopping-time decomposition (§8.2a) provides the correct analytical framework precisely because it bypasses the per-position transition matrices entirely.

Boundary-layer concentration. Approximately 95% of $R(K)$ comes from $\tau \leq 7$. The cumulative ratio through $\tau = 7$ at $K = 14$ is -2.66 , close to the full $R(14) = -2.62$. The cascade dynamics is overwhelmingly a boundary-layer phenomenon.

Even-odd alternation. The successive ratios $P(\tau + 1 \mid 00)/P(\tau \mid 00)$ exhibit a clear even-odd oscillation converging toward $1/2$: $\{0.254, 0.796, 0.783, 0.584, 0.664, 0.557, 0.626, \dots\}$. This is consistent with the Diaconis–Fulman spectral gap $\rho = 1/2$ as the asymptotic rate, modulated by the carry chain's non-stationary structure near the boundary.

8.3 The $\zeta(2)$ product (preliminary)

This claim is conditional on two unproved conjectures ($\alpha \rightarrow 1/6$ and $R \rightarrow -\pi$); the “new proof of the Basel problem” is speculative until both are established.

The combination of levels 1 and 2 with the conjectural $R = -\pi$ gives

$$\alpha \cdot R^2 = (1/6) \cdot \pi^2 = \zeta(2)$$

If both $\alpha = 1/6$ (Conjecture 2) and $R = -\pi$ (Conjecture 1) can be established analytically from carry arithmetic, this would yield a new proof of the Basel problem $\zeta(2) = \pi^2/6$ — one that begins with integer multiplication rather than Fourier analysis or contour integration.

Numerical verification. The product $\alpha(K) \cdot R(K)^2$ converges:

K	$\alpha \cdot R^2$	$ \alpha \cdot R^2 - \zeta(2) $
11	0.405	1.24
15	1.218	0.43
18	1.583	0.062
20	1.648	0.003
21	1.662	0.017

The product $\alpha \cdot R^2$ overshoots $\zeta(2)$ at $K = 21$ because α overshoots $1/6$ while R has not yet reached $-\pi$. The near-exact agreement at $K = 20$ (gap = 0.003) is partially accidental: it reflects a cancellation between the α overshoot and the R undershoot at that particular K .

8.4 Base specificity

The sub-sector ratio $\alpha = 1/6$ is **not universal** — it is specific to base 2. Monte Carlo experiments in bases 3, 5, and 7 show no convergence to $1/6$ or to any other Bernoulli number. In base 3, α is wildly unstable across K values; in base 5, the sector ratio itself appears rational ($R_5 \rightarrow 5/4$) with α_5 far from $1/6$. This base specificity makes the binary Bernoulli hierarchy more, not less, remarkable: it reflects a deep structural property of base-2 arithmetic, potentially connected to the ramification of 2 in $\mathbb{Z}[i]$ (Section 8.6). See [G] for the proof that base 2 is the unique base where the D-parity boundary is a straight line in angular coordinates.

8.5 The Euler product perspective

Fulman’s multiplicativity property $K_a \cdot K_b = K_{\{ab\}}$ for carry transfer operators [3] implies that the Markov spectral sum in base b factors as

$$S_b(0) = b^2/(b^2 - 1)$$

(verified numerically for $b = 2, 3, 5, 7, 10$). For $b = 2$ this gives $S_2(0) = 4/3$. (Note: this spectral zeta sum $S_2(0)$ is a different quantity from the Markov sector ratio $R_{\text{Markov}} = +2/3$ of Theorem 2; both derive from independent-convolution statistics but measure different observables.) Taking the product over all primes:

$$\prod_{\{p \text{ prime}\}} S_p(0) = \prod_p p^2/(p^2 - 1) = \zeta(2)$$

This is a restatement of the classical Euler product, but viewed through carry arithmetic: each prime p contributes its Markov spectral sum factor to $\zeta(2)$. The non-Markov (exact) sector ratio introduces $L(1, \chi_4) = \pi/4$ as the additional factor, connecting to the Dedekind zeta function (Section 8.6).

8.6 The Dedekind zeta function of $\mathbb{Q}(i)$ (speculative)

This is a heuristic connection: the factorization is classical; attributing both factors to carry arithmetic is conjectural and depends on Conjecture 1.

The Gaussian integers $\mathbb{Z}[i]$ have Dedekind zeta function

$$\zeta_{\{\mathbb{Q}(i)\}}(s) = \zeta(s) \cdot L(s, \chi_4)$$

The carry bridge produces: - $\zeta(2s)$ from the spectral zeta function (Theorem 3), via the Markov eigenvalue density - $L(1, \chi_4) = \pi/4$ from the sector ratio (Conjecture 1), via the effective χ_4 model (§6.2) and resolvent universality [E]

Both factors of $\zeta_{\{\mathbb{Q}(i)\}}$ thus emerge from the same arithmetic object — the carry bridge of binary multiplication. The prime 2 is the unique rational prime that *ramifies* in $\mathbb{Z}[i]$, satisfying $2 = -i(1+i)^2$. This ramification forces the character χ_4 into the spectral decomposition, explaining at the algebraic level why base 2 is unique [G]: for $b \geq 3$, the sector ratio appears to be rational (e.g., $R = 5/4$ for base 5), and the D-parity boundary is curved in angular coordinates, removing the geometric mechanism that produces $L(1, \chi_4)$.

9. Numerical Evidence

9.1 Exact enumeration

The sector ratio $R(K)$ is computed by exhaustive enumeration of all 4^{K-1} digit pairs. Enumeration through $K = 21$ covers 5.5×10^{11} pairs for $K = 21$ (C implementation with 10-thread parallelism, ~ 2.7 hours) [5].

K	$R(K)$	$ R(K) + \pi $	Sig. digits
7	-0.092	3.050	—
9	-0.726	2.416	—
11	-1.554	1.588	—
13	-2.339	0.803	0.1
15	-2.824	0.317	0.5
17	-3.035	0.107	1.0
19	-3.110	0.032	1.5
20	-3.125	0.017	1.8
21	-3.133	0.008	2.1

All values from the cascade valuation (Definition 1), cross-validated against [E, Table 1].

9.2 Richardson extrapolation

The raw convergence $R(K) \rightarrow -\pi$ is slow: at $K = 21$ we have 2.1 significant digits. Richardson extrapolation dramatically improves precision.

Ansatz 1 (polynomial). A natural first guess is $R(K) = R(\infty) + a_1/K^2 + a_2/K^4 + \dots$, motivated by the $O(1/L^2)$ finite-size correction in the Möbius sum $S(A, L)$ (Proposition 2). However, the true convergence rate is exponential (§9.3), and the polynomial Neville table diverges in practice.

Ansatz 2 (exponential). The empirical gap ratios $g(K)/g(K-1)$ converge monotonically to $1/2$ (§9.3, Table 1 of [E]), consistent with

$$R(K) = R(\infty) + c_1 \rho^K + c_2 \rho^{2K} + c_3 \rho^{3K} + \dots, \quad \rho = 1/2.$$

The Neville–Aitken Richardson table with step-size variable $h = \rho^K$ eliminates successive error terms.

Proposition 3 (Richardson extrapolation). Applying the Neville–Aitken deferred approach to the limit [9] to the sequence $R(K)$ for $K = 7, 9, 11, \dots, 21$ with Ansatz 2 ($h = (1/2)^K$) yields

$$R_{\text{extrap}} = -3.1419\dots \text{ (4.0 significant digits)}$$

using exact $R(K)$ for $K \leq 13$ and $K \geq 19$, with $K = 14\text{--}18$ approximations from the high-precision profiles of [E]. Using only exact values ($K \leq 13, K \geq 19$) the estimate improves to 4.4 significant digits. Leave-one-out cross-validation confirms stability: removing any single $K \in \{7, \dots, 17\}$ changes the estimate by less than 10^{-4} .

Remark (one-step check). The single Richardson step using only $K = 20$ and $K = 21$ gives $2R(21) - R(20) = -3.14194$, a gap of 3.5×10^{-4} from $-\pi$ (3.5 significant digits). This one-step estimate is consistent with the Neville-Aitken result, confirming that the full table is not needed to establish 3+ digit precision.

9.3 Convergence rate

The error $\varepsilon(K) = |R(K) + \pi|$ satisfies

$$\varepsilon(K) = O(K^2 \cdot 2^{-K})$$

for $K \geq 11$. The consecutive gap ratio $g(K)/g(K-1)$ forms a monotone sequence converging to $1/2$ from above: 1.000, 0.954, ..., 0.522, 0.491 at $K = 21$ (see [E, Table 1]). The gap ratio crossing 2 at $K = 21$ confirms the asymptotic regime. This exponential rate is consistent with the Diaconis-Fulman spectral radius $\rho = 1/2$ governing the carry chain [4].

Remark. A polynomial fit $\varepsilon \sim C/K^2$ provides a reasonable approximation for moderate K but misses the asymptotic structure. The exponential characterization $\varepsilon \sim C \cdot (1/2)^K$ captures the true rate. The Richardson extrapolation of §9.2 remains effective because at moderate K the polynomial and exponential forms are numerically similar.

9.4 Verification of the closed form

Theorem 1 is verified numerically at 50-digit precision using arbitrary-precision arithmetic. The algebraic identity $4r/(\gamma(1-r^2)) = 2\beta/(1+2\beta)$ holds to full precision for all tested values of β . The formula $S(\beta^*, L) \rightarrow -\pi$ is confirmed at $L = 1000$ with $|S + \pi| < 4 \times 10^{-5}$, consistent with the $O(1/L^2)$ rate.

9.5 Topological stability of minimum positions

The $K \rightarrow \infty$ limit is not only well-defined for $R(K)$ as a scalar but also for the positional structure of the carry object on the critical line. Comparing the mixed channel $|\mu_{\chi_4}(1/2 + it)|$ at $K = 21$ and $K = 999$ (Richardson proxy) on a grid with spacing $dt = 0.05$, all 15 tested local minima have positional shift = 0.000: no minimum moves by a single grid point under the $K = 21 \rightarrow 999$ extrapolation. The *values* of the minima are approximately flat across this range (§5.3 of [L]), and now the *positions* are confirmed stable to within the grid resolution.

This stability result means the zero-tracking behavior documented in §5.3 of [L] is not an artifact of the truncation at $K = 21$ but is already present in the $K \rightarrow \infty$ limit. It provides a structural answer to the natural objection that large- K behavior might shift the positions of the carry-amplitude minima relative to the L -function zeros.

10. The Gap: From S(A) to R(K)

10.1 What is proved

Theorem 1 establishes that the abstract alternating sum $S(A) = 2(1-A)/(3-2A)$ is a closed-form Möbius transformation. Setting $S = -\pi$ yields $A^* = (3\pi+2)/(2(\pi+1))$. The spectral zeta function $\zeta_F(s) \rightarrow \zeta(2s)$ is proved (Theorem 3). The Dirichlet character $\chi_4(n) = \sin(n\pi/2)$ emerges as the effective spectral weight (§6.2).

10.2 The gap and the universality phenomenon

The gap between the abstract sum $S(A)$ and the physical sector ratio $R(K)$ is the central open problem. Three observations constrain its nature:

Observation 1 (Distinct convergence rates). The abstract sum converges polynomially: $S(A, L) - S(A, \infty) = O(1/L^2)$. The physical sector ratio converges exponentially: $R(K) - R(\infty) = O(K^2/2^K)$.

Observation 2 (A_fit does not converge). Defining $A_{\text{fit}}(K)$ by the equation $S(A_{\text{fit}}, 2K-1) = R(K)$, the sequence $A_{\text{fit}}(K)$ does not converge to A^* .

Observation 3 (Carry projection is exact Markov). Projecting the exact (non-Markov) transfer operator onto carry degrees of freedom recovers the Markov transfer matrix exactly. The parameter A is therefore *not* a perturbative eigenvalue shift.

These observations rule out deriving A^* from the Fejér kernel correlation structure. Nonetheless, both $S(A, L)$ and $R(K)$ converge numerically to the same value $-\pi$ — a *universality* phenomenon [P2] that we state as a conjecture:

Conjecture 3 (Universality). Let $S(A^*, L) = 2(1 - A^*)/(3 - 2A^*)$ be the abstract Möbius sum (Theorem 1), and let $R(K) = \sigma_{10}(K)/\sigma_{00}(K)$ be the cascade sector ratio (Definition 2). Then

$$\lim_{L \rightarrow \infty} S(A^*, L) = \lim_{K \rightarrow \infty} R(K) = -\pi,$$

where the left equality is proved (Theorem 1) and the right is Conjecture 1. The two systems share different microscopic dynamics (polynomial vs exponential finite-size corrections) but the same macroscopic limit.

The common mechanism is the Dirichlet character χ_4 and the L-function $L(1, \chi_4) = \pi/4$. [E] identifies **resolvent universality** as the specific mechanism: the spectral resolvent

$$(I - \mathcal{K}_{\text{eff}})^{-1}$$

, applied to the true per-sector cascade profiles, produces a macroscopic sum converging to $-\pi$ through collective mode weighting. The left limit is an exact algebraic evaluation; the right is supported by 4.0-digit exponential Richardson extrapolation through $K = 21$ (§9).

10.3 Structural decomposition

Proposition 4 (Top-carry constraint). For a D -odd product $n = p \cdot q$ with $D = 2K-1$, the convolution at position $D-2$ equals exactly $\text{conv}_{D-2} = a + c$, where $a = p_{K-2}$ and $c = q_{K-2}$ are the sector bits.

Proof. $\text{conv}_{D-2} = p_{K-2} \cdot q_{K-1} + p_{K-1} \cdot q_{K-2} = a \cdot 1 + 1 \cdot c = a + c$. $\square$

Corollary. The D -odd condition requires $\text{carries}[D-1] = 0$, hence $(a + c + \text{carries}[D-2]) < 2$. For sector $(1,0)$: $a + c = 1$, so $\text{carries}[D-2] = 0$, forcing $M \leq D-3$. For sector $(0,0)$: $\text{carries}[D-2] \in \{0,1\}$, yielding two sub-populations. This gives $\sigma_{00} = \sigma_{00}^{\text{top}} + \sigma_{00}^{\text{low}}$, while $\sigma_{10} = \sigma_{10}^{\text{low}}$ has no top contribution. [E, §8.5] independently establishes this structural exclusion: $P(j=2|\text{sector } 10) = 0$, confirmed by backward dynamic programming ([E]).

Proposition 5 (Exact symmetry). $\sigma_{10}(K) = \sigma_{01}(K)$ for all K .

Proof. The substitution $(p,q) \mapsto (q,p)$ maps sector $(1,0)$ to $(0,1)$, preserving all carries. $\square$

Theorem 4 ($N_{11} = 0$). For all $K \geq 2$, no D -odd pair has sector $(1,1)$.

Proof. In sector $(1,1)$, $\text{conv}_{D-2} = 2$. The D -odd condition requires $\text{carries}[D-1] = 0$, but $(2 + \text{carries}[D-2])/2 \geq 1$, contradiction. $\square$

Theorem 5 (Closed-form count ratio). In the continuous limit $K \rightarrow \infty$,

$$N_{10}/N_{00} \rightarrow \frac{2 \ln(4/3) - 1/2}{2 \ln(9/8)} = 0.31992784225 \dots$$

with correction $O(1/2^K)$.

Proof. Write $X = 2^{K-1}(1+u)$, $Y = 2^{K-1}(1+v)$ with $u, v \in [0, 1)$. The D -odd condition $(1+u)(1+v) < 2$ defines a region in (u,v) . Sectors are determined by $\lfloor 2u \rfloor$ and $\lfloor 2v \rfloor$: area $f_{00} = 2 \ln(9/8)$, $f_{10} = 2 \ln(4/3) - 1/2$, $f_{11} = 0$. Verified to 8 significant digits against exact enumeration ($K = 3-21$). $\square$

Corollary (π isolation). Since $R = (\langle \text{val} \rangle_{10} / \langle \text{val} \rangle_{00}) \times (N_{10}/N_{00})$ and N_{10}/N_{00} involves only $\ln 2$ and $\ln 3$, the entire factor π enters through the carry weight ratio:

$$\langle \text{val} \rangle_{10} / \langle \text{val} \rangle_{00} \rightarrow -\pi \cdot \frac{2 \ln(9/8)}{2 \ln(4/3) - 1/2} = -9.8197 \dots$$

Any proof must explain how the sector bit perturbation modifies the *distribution* of c_{M-1} to produce this ratio. The Dirichlet series approach (character decomposition by $n \bmod 4$) does not cleanly factor R , since $\sim 75\%$ of D -odd products are even.

Remark. [E, §9.2] falsifies the hypothesis that the *cascade count* ratio n_{00}/n_{10} converges to π ; the limit is algebraic ($\approx 25/8$). Combined with Theorem 5, this confirms that π is entirely *dynamical*: it resides in the carry weight functional, not in any geometric count.

10.4 The carry weight mechanism

The total variation distance $TV(j)$ between the carry distributions in sectors (1,0) and (0,0) reveals a boundary layer with two peaks: $TV \approx 0.16$ at $j = K-2$ (sector bit perturbation) and $TV \approx 0.54$ at $j = D-2$ (top structural constraint), with a plateau $TV \approx 0.06$ in between.

Four steps link the sector bit to the carry weight $val = c_{\{M-1\}} - 1$:

1. **Convolution perturbation (position K-2).** The sector bit $a = 1$ adds $q_{\{j-K+2\}}$ to $conv_j$ for $j \in [K-2, 2K-3]$, generating additional carries in the bulk.
2. **Carry propagation.** The extra carries propagate upward through $\sim K$ positions; the mean carry at each position is ~ 0.35 higher in sector (1,0) [F].
3. **Top constraint (position D-2).** The D-odd condition forces $carries[D-2] = 0$ in sector (1,0), eliminating the “top” population and shifting M to $D-3$ or lower.
4. **Val distribution shift.** The higher carry levels in sector (1,0), combined with the constrained top, produce 30% with $c_{\{M-1\}} = 2$ ($val = +1$) vs 5.5% in sector (0,0){low}, driving $\sigma_{\{10\}} > 0$ while $\sigma_{\{00\}} < 0$.

Spectral connection. The sector bit perturbation is a ramp (constant $\langle \Delta conv_j \rangle = 1/2$) over half the chain. In the bridge eigenmode basis $\sin(n\pi j/L)$, the ramp Fourier coefficients $\Delta v_n = [\cos(n\pi/2) - \cos(n\pi)]/(n\pi)$ combined with the source-point factor $\sin(n\pi/2)$ satisfy:

$$\sin(n\pi/2) \cdot [\cos(n\pi/2) - \cos(n\pi)] = \sin(n\pi/2) \text{ if } n \text{ is odd, and } 0 \text{ if } n \text{ is even.}$$

The even-mode contributions cancel exactly; the odd-mode sum becomes $\sum \sin(n\pi/2)/n = \pi/4 = L(1, \chi_4)$. This explains conceptually how the Leibniz series enters the problem.

Remaining normalization gap. With the Diaconis-Fulman spectral gap $\lambda = 1/2$ taken as constant, the spectral sum evaluates to $1/2$, not $-\pi$. The discrepancy factor -2π must originate from: (i) the mode-dependent eigenvalue spectrum of the non-Markovian carry chain; (ii) the non-local observation functional $val = c_{\{M-1\}} - 1$ evaluated at the random stopping position $M-1$; and (iii) the redistribution of the $M = D-2$ population ($\sim 55\%$ of sector (0,0)) into $M \leq D-3$ in sector (1,0). All three concern carry weights, consistent with the π -isolation corollary. The analytical resolution is identified in [E, §9] as the **Holte bridge / Doob h-transform** matching between the bulk carry distribution (conditioned on D-odd) and the boundary layer exclusions.

10.5 Analytical evidence: continuum limit and exact symbolic forms

Carry recursion from MSB. Define $carry[d] = \text{carry at position } D-1-d \text{ from below}$, with recursion $carry[d] = 2 \cdot carry[d-1] + b_d - C_d$ and $carry[0] = 0$.

Theorem 7 (Unit leading carry). Let d_0 be the smallest $d \geq 1$ with $\text{carry}[d] \geq 1$. Then $\text{carry}[d_0] = 1$ exactly.

Proof. Since $\text{carry}[d_0 - 1] = 0$, $\text{carry}[d_0] = b_{d_0} - C_{d_0}$. For $\text{carry} \geq 1$ with $b_{d_0} \in \{0, 1\}$ and $C_{d_0} \geq 0$: necessarily $C_{d_0} = 0$ and $b_{d_0} = 1$. $\square$

Corollary (Universal val formula). $\text{val} = 1 + b_{d_0+1} - C_{d_0+1}$ for all depths d_0 and both sectors.

The continuum function $\text{val}_\infty(u, v) = \lim_{K \rightarrow \infty} \text{val}(u, v; K)$ is a well-defined step function taking values in $\{-1, 0, +1\}$, giving sector sums $\sigma_{ac} = \iint_{\text{sector } ac, \text{D-odd}} \text{val}_\infty(u, v) du dv$.

Exact symbolic forms. Each $\sigma(d_0)$ admits an exact closed form as a rational linear combination of $\{1, \ln p : p \text{ prime}, p \leq 2^{d_0+2}\}$:

$$\sigma_{00}(1) = \frac{1}{2} + \frac{9}{2} \ln 2 - 2 \ln 3 - 3 \ln 5 + \frac{7}{4} \ln 7$$

The individual coefficients grow as $\sim 4^{d_0}$ while $\sigma(d_0) = O(2^{-d_0})$. After 7 depth levels, the cumulative $\ln 2$ coefficient reaches ~ 1523 , while the total $\sigma_{00} \approx -0.007$ — catastrophic cancellation (ratio $> 10^5$) essential to the emergence of π . At each depth d_0 , new primes in $[2^{d_0+1}, 2^{d_0+2}]$ enter, so the infinite series involves **all primes**. By Baker’s theorem, no finite combination of logarithms of algebraic numbers equals π ; the infinite series circumvents this obstruction.

$d_{\max}$	σ_{10}/σ_{00}	$ R + \pi $	digits
5	-2.695	4.5×10^{-1}	0.85
10	-3.098	4.3×10^{-2}	1.86
12	-3.129	1.2×10^{-2}	2.41
14	-3.139	2.1×10^{-3}	3.18
15	-3.14146	1.4×10^{-4}	4.36

Beyond $d_0 = 15$, catastrophic cancellation exceeds float64 precision. Extended precision (60 digits; [E]) reaches $d_0 = 18$ but reveals a structural limitation: the depth-first carry-free enumeration captures only 17-33% of the D-odd configuration space, causing the ratio to worsen beyond $d_0 = 15$. This is a method limitation, not a failure of the $-\pi$ conjecture.

11. Open Problems

1. Prove $\sigma_{10}/\sigma_{00} = -\pi$ analytically. The decomposition $R = R_0 + \Delta R$ (where $R_0 = -3.9312\dots$ involves only $\ln 2$ and $\ln 3$, and $\Delta R = +0.7896\dots$ carries the entire transcendental content) reduces the problem to showing that the cascade correction $\Delta R = |R_0| - \pi$. Six equivalent formulations are catalogued in §10.5; the depth series $\sum_{d=1}^{\infty} \sigma(d)$ converges to 4.36 significant digits at $d_{\max} = 15$. The most promising analytical path is the **stopping-time series** (§8.2a): derive $P(\tau \mid ab)$ and $E[\text{val} \mid \tau, ab]$ analytically from the carry chain structure, then sum $R = \sum_{\tau} [n_{10}(\tau) \cdot E[\text{val} \mid \tau, 10] + \dots]$. The sector asymmetry $\Delta(\tau) = E[v \mid \tau, 10] - E[v \mid \tau, 00]$, which converges to K -independent constants (Table 1, §8.2a), is the core quantity: it encodes the entire transcendental content of R . The per-position spectral approach (Doob h-transform of local transition matrices) is ruled out: §8.2b shows that the per-position eigenvalues grow with K instead of stabilizing, so the LMH must operate at the global resolvent level, not locally. The companion analysis of [E, §9] reduces the remaining closure to two explicit points: a scalar $\chi_4 \rightarrow L(1, \chi_4)$ constant (data-constrained to $C \in [-4.0032, -3.9896]$) and an analytic proof of the dominant $1/2$ decay rate under D-odd conditioning. *Computational status.* Direct enumeration reaches $K = 21$. The single Richardson step $2R(21) - R(20) = -3.14194$ (§9.2, experiment P1_08) confirms 3.5 significant digits; the full Neville-Aitken table (§9.2, experiment P1_07) yields 4.0 digits. The $K = 999$ carry ensemble provides independent spectral confirmation: the χ_4 -weighted minima track the L-zeros at $K = 999$ to the same positional accuracy as at $K = 21$ (§9.5, experiment P1_08; [L, §5.5]). The computational problem is resolved to 4+ digits; the analytical proof remains open.
2. **Base universality.** For base $b \geq 3$, the Markov sum $S_b(0) = b^2/(b^2-1)$. Is the exact sector ratio R_b rational for all $b \geq 3$? Exact enumeration for base 3 and base 5 would test this. The conjecture $R_5 = 5/4$ has preliminary numerical support.
3. **Higher Bernoulli levels.** The hierarchy $B_0, |B_1|, B_2$ is confirmed through Level 2. Level 3 ($B_3 = 0$) is proved by Theorem 6: the cascade rigidity forces $\sigma_{00}^{c_2 \geq 1, c_3 = 1} = 0$ exactly, and the recursive extension produces the same vanishing at every depth. The Level 4 signal $|B_4| = 1/30$ does not appear as a simple partition-function ratio, asymptotic correction coefficient, or spectral gap (all tested through $K = 13$). The non-trivial dynamics telescopes into the $(c_2, c_3, \dots) = (0, 0, \dots)$ sub-population at each level. Identifying the correct Level 4 observable remains open.
4. **Catalan's constant from carries.** The value $L(2, \chi_4) = G = 0.9159\dots$ (Catalan's constant) is the next L-function value after $L(1, \chi_4) = \pi/4$. Can it be extracted from carry chain data?
5. **$\zeta(s)$ at odd arguments** (speculative). The Dedekind factorization $\zeta(s) = \zeta_{\{\mathbb{Q}(i)\}}(s)/L(s, \chi_4)$ gives $\zeta(s)$ at all arguments if $L(s, \chi_4)$ is accessible from carries for general s . The case $\zeta(3) = 1.20206\dots$ (Apéry's constant) is the most important test.

6. **New proof of the Basel problem** (conditional). If both $\alpha = 1/6$ (Conjecture 2) and $R = -\pi$ (Conjecture 1) are proved from carry arithmetic, then $\zeta(2) = \alpha R^2 = \pi^2/6$ follows. This depends on two currently unproved conjectures; neither is close to resolution.
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